Layer Consensus Decoding: When Internal Disagreement Fails to Predict Hallucinations

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Abstract

Large language models frequently generate plausible but factually incorrect outputs (hallucinations), limiting their deployment in high-stakes applications. We investigate Layer Consensus Decoding (LCD), a training-free approach that attempts to prevent hallucinations by monitoring agreement between predictions from different transformer layers during generation. The hypothesis is that correct generations exhibit consistent predictions across layers, while hallucinations show layer-wise disagreement. We compute consensus scores at each decoding step by comparing probability distributions from early and late layers using Jensen-Shannon divergence. Experiments on TriviaQA, BoolQ, and CommonsenseQA using Mistral-7B reveal that layer consensus signals fail to reliably distinguish correct from hallucinated generations, achieving near-random AUROC scores (0.43–0.52). Despite moderate hallucination prevention rates, extensive ablations testing vocabulary overlap, concentration variance, and layer selection strategies show no consistent improvement. Our negative results highlight fundamental challenges in using layer-wise agreement as a hallucination indicator and suggest that more sophisticated features or task-specific calibration may be necessary.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across diverse tasks, yet they frequently generate plausible but factually incorrect content—a phenomenon known as hallucination (**?**). This undermines their reliability in high-stakes domains such as medical diagnosis (**?**), legal reasoning, and factual question answering. While post-hoc detection methods exist (**??**), they cannot prevent the cascade effect where early uncertain tokens lead to increasingly confident but incorrect continuations.

Recent work has explored layer-wise analysis for understanding LLM behavior. The Tuned Lens (**?**) enables visualization of predictions at intermediate layers, while studies show that uncertainty affects layer-wise inference dynamics (**?**). CROW (**?**) identifies layer inconsistency in backdoored models, and Active Layer-Contrastive Decoding (ActLCD) (**?**) uses reinforcement learning to learn when to contrast layers for improved factuality.

We investigate whether layer-wise prediction disagreement can serve as a training-free indicator of hallucinations. Our hypothesis is that correct generations should show smooth transitions across layers, while hallucinations exhibit instability as early layers express uncertainty that later layers mask. We compute consensus scores using Jensen-Shannon divergence between early and late layer predictions during generation. Our experiments on three question-answering benchmarks using Mistral-7B reveal that this hypothesis does not hold in practice. Layer consensus scores show heavy overlap between correct and hallucinated outputs, achieving AUROC values near random chance (0.43–0.52). Extensive ablations testing alternative metrics, layer selection strategies, and per-token trajectory analysis consistently fail to improve separability.

2 Related Work

Layer-wise Analysis. The Tuned Lens (**?**) enables decoding intermediate layer representations into vocabulary distributions, revealing how predictions evolve across layers. Kim et al. (**?**) demonstrate that epistemic uncertainty affects layer-wise probability trajectories differently for correct versus incorrect predictions. However, these works focus on post-hoc analysis rather than real-time hallucination prevention.

Hallucination Detection. Semantic entropy (?) detects hallucinations by measuring uncertainty at the meaning level through multiple sampled generations. Token-level uncertainty methods (?) quantify confidence in specific claims using final-layer probabilities. These approaches operate post-hoc and require either multiple forward passes or auxiliary models, unlike our single-pass layer consensus approach.

Decoding Strategies. Contrastive Decoding (?) improves generation by contrasting expert and amateur models, while self-consistency (?) aggregates multiple reasoning paths through majority voting. ActLCD (?) uses reinforcement learning to learn when to apply layer contrasting. Unlike these methods requiring training or multiple generations, we explore whether layer disagreement alone provides useful signals without any training.

3 Method

Layer-wise Prediction Extraction. For a transformer with L layers, we extract hidden states at layers $[\frac{L}{4}, \frac{L}{2}, \frac{3L}{4}, L]$ during each decoding step. We project each layer's final token representation through the language model head to obtain logits, then apply softmax to get probability distributions P_{early} (averaged over layers $\frac{L}{4}$ and $\frac{L}{2}$) and P_{late} (averaged over layers $\frac{3L}{4}$ and L).

Consensus Score Computation. We compute the Jensen-Shannon (JS) divergence between P_{early} and P_{late} as our primary consensus metric:

$$\text{JS}(P_{\text{early}}, P_{\text{late}}) = \frac{1}{2} D_{\text{KL}}(P_{\text{early}} \| M) + \frac{1}{2} D_{\text{KL}}(P_{\text{late}} \| M) \tag{1}$$

where $M = \frac{1}{2}(P_{\text{early}} + P_{\text{late}})$ and D_{KL} is the Kullback-Leibler divergence. Lower JS divergence indicates higher agreement between layers. We aggregate per-token scores by averaging across all generated tokens for each sample.

Alternative Metrics. We explore three additional consensus metrics: (1) Top-1 agreement: binary indicator of whether early and late layers predict the same top token; (2) Rank correlation: Spearman correlation between probability rankings of the top-100 tokens; (3) Entropy gap: difference in prediction entropy between late and early layers, capturing confidence shifts.

Hallucination Labeling. For each dataset, we label generations as correct (0) or hallucinated (1) by comparing generated answers to ground truth. For TriviaQA (?), we check if any answer alias appears in the generation. For BoolQ (?), we verify yes/no matches. For CommonsenseQA (?), we check if the correct option letter appears in the first few tokens.

4 Experimental Setup

Models and Datasets. We use Mistral-7B-Instruct (?), a 7B parameter instruction-tuned model. We evaluate on three question-answering benchmarks: TriviaQA (150 samples), BoolQ (150 samples), and CommonsenseQA (150 samples). These datasets span factual recall, boolean reasoning, and commonsense knowledge.

Evaluation Metrics. We measure hallucination detection performance using AUROC (Area Under ROC Curve) and AUPRC (Area Under Precision-Recall Curve). We compute Hallucination Prevention Rate (HPR) at multiple threshold percentiles. Task accuracy measures the percentage of correct answers.

Ablation Studies. We conduct ablations on vocabulary overlap (measuring top-k token set intersection, $k \in \{5, 10, 20, 50\}$), layer selection (quartile-based versus adjacent layer pairs), concentration variance (variance in probability mass concentration), token position analysis, and early spike detection. Additional ablations are provided in the appendix.

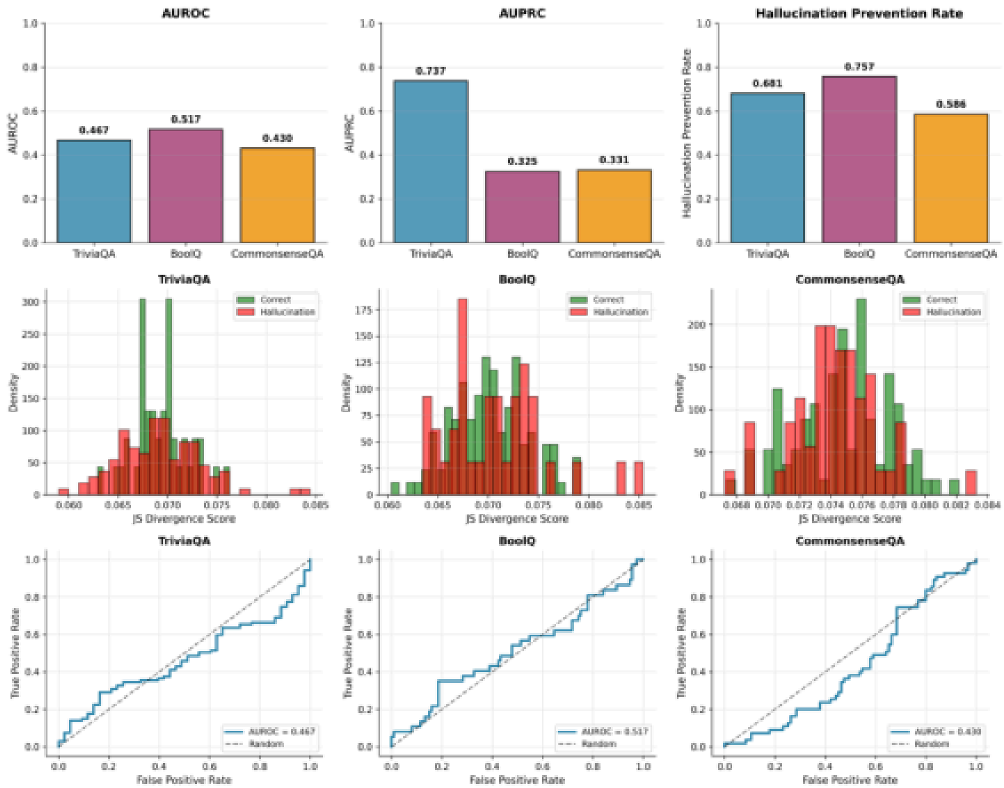

Figure 1: **Main performance metrics and distributions.** Top: AUROC values cluster around chance level (0.43–0.52) across all three datasets, indicating poor separability. AUPRC is inflated by class imbalance on TriviaQA (71% hallucination rate). Bottom: JS divergence score distributions show heavy overlap between correct (green) and hallucinated (red) cases. ROC curves track close to the diagonal, confirming near-random discrimination performance.

5 Results

Layer Consensus Fails to Predict Hallucinations. Figure 1 presents our main results. AUROC scores are near random chance across all three datasets: TriviaQA (0.467), BoolQ (0.517), and CommonsenseQA (0.430). This indicates that JS divergence between early and late layers does not reliably separate correct from hallucinated generations. The AUPRC for TriviaQA (0.737) appears high but is driven by class imbalance (71% hallucination rate) rather than strong ranking ability. The consensus score distributions exhibit substantial overlap between correct and hallucinated samples across all three datasets, with scores concentrated in a narrow band (0.06–0.09). The ROC curves closely follow the diagonal reference line, visually confirming the poor discrimination ability. These results directly challenge the hypothesis that hallucinations manifest as layer disagreement—both correct and incorrect generations show similar consensus patterns.

Token-Level and Alternative Metrics Show High Variability. Figure 2 (left) plots per-token consensus scores across generation steps for two representative datasets. Both correct and hallucinated samples show substantial within-class variability, with trajectories frequently crossing. Some hallucinations start with low divergence and remain stable, while others spike mid-sequence. Conversely, correct generations sometimes exhibit large divergence spikes. This inconsistency undermines the hypothesis that hallucinations manifest as sustained layer disagreement during generation. The temporal dynamics reveal that layer consensus is not a stable indicator—it fluctuates substantially within individual generations regardless of correctness. Figure 2 (right) compares four consensus metrics. Top-1 agreement is too binary and sparse, with most values at zero regardless of correctness. Rank correlation clusters tightly around zero for both classes, offering no discriminative power. The entropy gap (late minus early layer entropy) shows all negative values with minimal class separation.

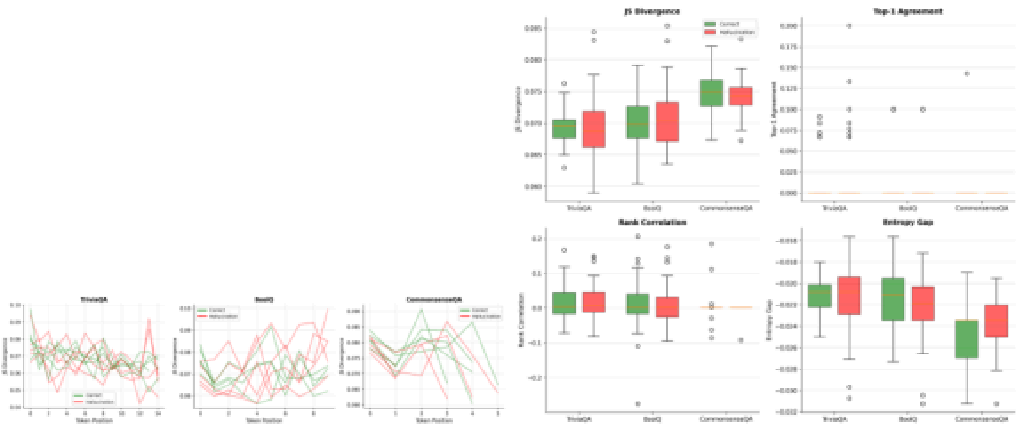

Figure 2: **Per-token trajectories and alternative metrics.** Left: Per-token consensus trajectories showing JS divergence across generation steps for TriviaQA and BoolQ. Lines are highly intertwined with frequent crossings between correct (green) and hallucinated (red) generations, showing no consistent pattern. Right: Comparison of four consensus metrics. All metrics show substantial overlap between correct and hallucinated distributions. Top-1 agreement is too sparse, rank correlation clusters near zero, and entropy gap shows minimal separation.

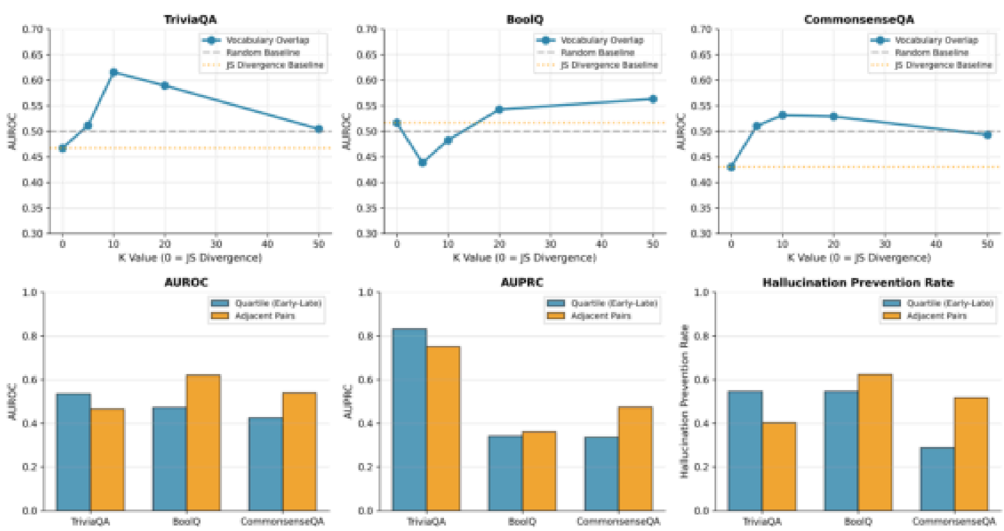

Figure 3: **Ablation studies on vocabulary overlap and layer selection.** Top: Vocabulary overlap ablation testing top-k token intersection size. Performance varies inconsistently across datasets and k values: TriviaQA peaks at k=10 (AUROC 0.61), BoolQ shows a U-shape with minimum at k=5, and CommonsenseQA improves gradually. No configuration works well across all datasets. Bottom: Comparison of layer selection strategies. Quartile-based selection (comparing layers $\frac{L}{4}$, $\frac{L}{2}$ against $\frac{3L}{4}$, L) and adjacent layer pairs show no consistent performance difference across datasets or metrics.

These results suggest that the problem is not specific to JS divergence but rather fundamental to using layer agreement as a hallucination signal. The failure extends across multiple mathematical formulations of consensus.

Ablation Studies Show No Improvement. Figure 3 (top) shows results for vocabulary overlap metrics. Instead of measuring divergence over the full vocabulary, we compute the intersection size of the top-k tokens predicted by early and late layers. Performance varies inconsistently: TriviaQA shows a peak at k=10 (AUROC 0.61), BoolQ exhibits a U-shape with minimum at k=5, and CommonsenseQA shows gradual improvement plateauing around k=20. However, no single configuration works well across all datasets, and the absolute AUROC values remain modest. This dataset-dependent behavior suggests that vocabulary overlap captures task-specific artifacts rather than general hallucination patterns. Figure 3 (bottom) compares our default quartile-based layer selection with adjacent layer pairs. Neither strategy shows consistent superiority across datasets or metrics. This suggests that the fundamental issue is not about selecting the right layers to compare, but rather that layer-wise agreement itself is not a reliable hallucination indicator. We also tested concentration variance (measuring variance in probability mass rather than distribution similarity) and found similarly poor performance with heavy overlap between classes (see Appendix).

Summary of Negative Results. Across all experiments, we consistently observe: (1) Near-random AUROC scores (0.43–0.52) for the primary JS divergence metric; (2) Heavy distributional overlap between correct and hallucinated samples for all tested metrics; (3) Inconsistent and dataset-dependent patterns in ablation studies; (4) No layer selection or alternative metric strategy that consistently improves performance. The moderate HPR values (0.59–0.76) may reflect general conservative regularization rather than targeted hallucination detection, as they occur despite poor discrimination ability.

6 Discussion

Our results reveal fundamental challenges in using layer-wise prediction agreement as a hallucination indicator. **Global vs. Local Signals:** Computing divergence over the entire vocabulary may be too coarse-grained. Hallucinations might manifest in disagreement over specific token types (e.g., named entities, numbers) rather than overall distribution shifts. **Model-Dependent Behavior:** Our experiments use Mistral-7B-Instruct, an instruction-tuned model. Different architectures or training procedures might exhibit different layer-wise dynamics. However, the consistency of negative results across three diverse datasets suggests the problem is not dataset-specific. **Task Complexity:** The relationship between layer agreement and correctness may depend on task complexity and answer length. Short answers (BoolQ) versus long answers (TriviaQA) show similar patterns, suggesting length is not the primary factor. **Calibration Requirements:** While we tested multiple thresholds, the poor AUROC scores indicate that no threshold can reliably separate classes. This suggests the signal itself is weak rather than simply requiring better calibration. **Alternative Hypotheses:** Our results challenge the hypothesis that hallucinations consistently manifest as layer disagreement. Possible explanations include: (1) Both correct and hallucinated generations can exhibit layer disagreement during normal reasoning; (2) Hallucinations may involve confident agreement across layers when the model has learned incorrect associations; (3) Layer dynamics might reflect prompt/context processing rather than answer correctness.

7 Conclusion

We investigated whether layer-wise prediction consensus could serve as a training-free indicator of hallucinations in large language models. Our experiments on three question-answering benchmarks using Mistral-7B reveal that simple layer agreement metrics fail to reliably distinguish correct from hallucinated generations. JS divergence between early and late layers achieves near-random AUROC scores (0.43–0.52), with heavy distributional overlap between classes. Extensive ablations testing vocabulary overlap, concentration variance, alternative layer selections, and per-token trajectory analysis show no consistent improvement. These negative results provide important insights for the community. They suggest that layer-wise agreement alone is insufficient for hallucination detection and that more sophisticated approaches are needed. Future work might explore token-type-specific consensus, task-adaptive calibration, or combining layer signals with other uncertainty indicators. Our findings highlight the importance of reporting negative results to prevent duplicated effort and guide research toward more promising directions for improving LLM reliability.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTAL DETAILS

Computational Resources. All experiments were conducted on a single NVIDIA A100 GPU with 40GB memory. Each dataset evaluation (150 samples) took approximately 45–60 minutes. The model was loaded in float16 precision to fit within memory constraints.

Hyperparameters. We used greedy decoding with maximum generation length of 20 tokens for TriviaQA and CommonsenseQA, and 10 tokens for BoolQ. Temperature was set to 1.0 (no scaling). Layer indices were computed as $[\lfloor L/4 \rfloor, \lfloor L/2 \rfloor, \lfloor 3L/4 \rfloor, L-1]$ where L is the total number of layers.

Numerical Stability. We clipped probabilities to $[10^{-10}, 1.0]$ and renormalized before computing divergences to avoid numerical issues. JS divergence values were clipped to $[0, 1]$ and any infinite or NaN values were replaced with 0.5.

B ADDITIONAL ABLATION RESULTS

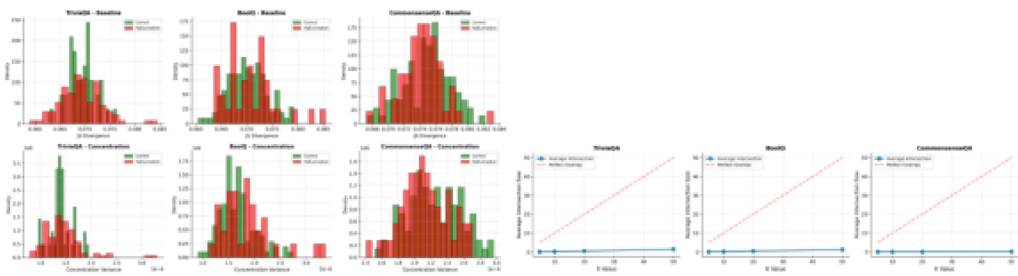

Figure 4: **Concentration variance and vocabulary intersection ablations.** Left: Concentration variance ablation measuring variance in probability mass concentration rather than distribution similarity. Top row shows baseline JS divergence distributions, bottom row shows concentration variance. Both metrics exhibit heavy overlap between correct and hallucinated cases, with concentration variance showing even tighter clustering. Right: Average vocabulary intersection size for different k values. The blue line (actual intersection) remains near zero (typically below 2 tokens) and far below the red reference line (perfect overlap), indicating minimal agreement between early and late layer top-k predictions.

Figure 4 (left) compares JS divergence with concentration variance (variance in the probability mass of top-k tokens across layers). The concentration metric shows even tighter clustering than JS divergence, with distributions heavily overlapping between correct and hallucinated samples. This suggests that measuring how probability mass concentrates across layers is no more informative than measuring distributional similarity. Figure 4 (right) shows that vocabulary intersection sizes remain minimal across all k values, typically below 2 tokens even when comparing top-50 predictions. This explains why vocabulary overlap metrics provide limited signal—early and late layers simply predict very different token sets, regardless of whether the final answer is correct or hallucinated.

C DETAILED PER-DATASET ANALYSIS

TriviaQA. With 28.67% accuracy and 71% hallucination rate, this was the most challenging dataset. The high AUPRC (0.737) is misleading—it primarily reflects the high base rate of hallucinations rather than effective ranking. The narrow JS divergence range (0.060–0.085) suggests the model's layer dynamics are similar regardless of answer correctness.

BoolQ. Despite 75.33% accuracy, AUROC (0.517) remains near chance. The yes/no format might be expected to show clearer layer consensus patterns, but we observe similar overlap to other datasets.

This suggests that even for simple binary decisions, layer agreement does not reliably indicate correctness.

CommonsenseQA. With 63.33% accuracy and the lowest AUROC (0.430), this dataset shows the weakest layer consensus signal. The multiple-choice format requires selecting among specific options, yet layer disagreement does not correlate with incorrect choices. This may reflect that commonsense errors involve confident but wrong reasoning rather than uncertain generation.